SNMF-Net: Learning a Deep Alternating Neural Network for Hyperspectral Unmixing

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***Abstract*— Hyperspectral unmixing is recognized as an important tool to learn the constituent materials and corresponding distribution in a scene. The physical spectral mixture model is always important to tackle this problem because of its highly ill-posed nature. In this article, we introduce a linear spectral mixture model (LMM)-based end-to-end deep neural network named SNMF-Net for hyperspectral unmixing. SNMF-Net shares an alternating architecture and benefits from both model-based methods and learning-based methods. On the one hand, SNMF-Net is of high physical interpretability as it is built by unrolling L_p sparsity constrained nonnegative matrix factorization (L_p-NMF) model belonging to LMM families. On the other hand, all the parameters and submodules of SNMF-Net can be seamlessly linked with the alternating optimization algorithm of L_p-NMF and unmixing problem. This enables us to reasonably integrate the prior knowledge on unmixing, the optimization algorithm, and the sparse representation theory into the network for robust learning, so as to improve unmixing. Experimental results on the synthetic and real-world data show the advantages of the proposed SNMF-Net over many state-of-the-art methods.**

***Index Terms*— Hyperspectral unmixing, model-based neural network, nonnegative matrix factorization (NMF), sparse representation.**

I. Introduction

HYPERSPECTRAL images (HSIs) capture not only the spatial information but also the spectral information indexed by numerous contiguous narrow bands of the object. These wealthy spectral bands offer fine details on the material information of the scene and dramatically augment HSIs discriminative ability. Benefiting from this, HSIs have enabled many practical applications, such as object recognition [1], object tracking [2], and medical analysis [3], [4], just to name a few. However, HSIs suffer from low spatial resolution caused by the limitation of hyperspectral sensors, leading to many "mixed pixels" containing more than one material. Hyperspectral unmixing aims to decompose every pixel into the combination of constituent materials, i.e., endmembers and their fractional abundances. It provides an attractive way to tackle this problem and greatly improves the utility of HSIs.

The unmixing problem can be generally considered as an ill-posed inverse problem recovering endmembers and abundances from acquired HSIs. Prior knowledge is very important to address this problem. Traditional model-based methods make elaborate hypotheses on the physical interaction between light and the target surface, i.e., the spectral mixture model, and, therefore, are highly interpretable. As a simplified model, the linear spectral mixture model (LMM) assumes the observed light interacts with only one material and linearly represents a pixel with endmembers [5]. Depended on LMM, the unmixing can be sequentially decomposed in two stages, i.e., endmember extraction, such as vertex component analysis (VCA) [6] and abundance estimation using regression, e.g., fully constrained least-squares (FCLS) [7]. Alternatively, this problem can be regarded as a blind source separation problem that simultaneously infers the endmembers and abundances. Formulating unmixing as a matrix factorization problem, the nonnegative matrix factorization (NMF) decomposes an HSI into the product of two factor matrices to perform unmixing [8]–[10]. Due to the inherent advantages of learning part-based representation of the data, the two matrices are, respectively, reasonably interpreted as endmembers and abundances in LMM. The nonconvex objective function of NMF makes its solution space very huge. Many extended NMFs embed several kinds of regularization terms encoding certain prior knowledge, such as sparsity, spatial information, and spatial–spectral information into the NMF model to limit the number of trivial solutions [11]–[17]. Instead of a 2-D matrix, nonnegative tensor factorization (NTF) directly treats an HSI as a third-order tensor for unmixing [18], [19]. Albeit interpretability, the model-based methods entail an accurate model depicting the unmixing problem, which is actually difficult to obtain. In addition, burdensome hyperparameter setting and computationally demanding iterations also restrict their effectiveness and efficiency.

Instead of model-based unmixing, a vast majority of learning-based particularly deep neural network (DNN)-based methods directly learn the hidden concepts by feeding a large number of training samples [20], [21]. In this way, learning-based methods are very fast. Blind unmixing can be

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cast as an unsupervised learning problem that jointly learns endmembers and abundances from data. Autoencoders (AEs) are mostly studied among these methods. AE comprises an encoder component mapping the input data into hidden concepts, i.e., abundances, and a decoder component reconstructing the input data by the learnable base, i.e., endmembers. Specifically, Su *et al.* [22] employed variational AE (VAE) to perform unmixing and stacked denoising AEs to generate favorable initialization for subsequent VAE. Instead of the Euclidian distance, EndNet used spectral angle distances (SADs) in the loss function to more consider the nonlinearity of endmembers [23]. Min *et. al* [24] further introduced the Wasserstein distance as a regularization term to better consider the distribution similarity between the observation and the reconstruction. Under the observation that the abundance maps of different endmembers are nearly orthogonal, Dou *et al.* [25] embedded orthogonal sparse prior into the AE to better consider the relationship among abundance maps. Qu and Qi [26] introduced an untied denoising AE (uDAS) that decouples the decoder from the encoder weights for more accurate endmember extraction. Moreover, cube-based convolutional AEs exploit the spatial information along with information of HSIs for spectral–spatial unmixing in which the encoder component is characterized by several convolutional layers [27], [28]. In addition to AE-based unmixing, Zhao *et al.* [29] and Rasti *et al.* [30] learned the deep priors of the abundance maps rather than handcrafted priors and integrated them in the unmixing framework to boost the unmixing quality. Without spectral mixture model, there is, however, still no human-interpretable way to understand why data-driven DNNs can perform unmixing, i.e., they are "black box" models. Their overparameterized architecture always demands many training samples. In practice, the high cost of collecting abundant HSIs and difficulties of obtaining ground-truth endmembers and abundances can also make the learning infeasible, dramatically hindering the generalization capability [31].

Complementary to model-based and data-driven learning-based methods, model-based DNNs embed physical models into the design of deep architectures through the deep unrolling technique. By introducing some learnable parameters, deep unrolling maps each iteration of model-based methods into a typical layer of DNNs and stacks a predefined number of layers to form a hierarchical deep network architecture. The unrolled networks naturally inherent advantages from two kinds of approaches, such as high interpretability, good generalization capability, strong learning ability, and computational efficiency, overcome the difficulties of modeling the underlying physical mechanism, and need much fewer training samples. Due to the above merits, deep unrolled networks have been successfully applied in many practical applications, such as image supperresolution [32], [33], rain removal [34], image deblurring [35], [36], and semantic segmentation [37], [38].

Recently, Qian *et. al* [39] have introduced an L_1-NMF-based NN for blind unmixing (MNN-BU). MNN-BU is obtained from LMM and the corresponding iterative shrinkage-thresholding algorithm (ISTA). It shares an AE structure. The encoder component is obtained by unrolling ISTA, and the decoder component is an additional learnable endmember matrix set behind the encoder. The experimental results demonstrate that MNN-BU outperforms several baselines in terms of both efficiency and accuracy. Despite this, we analyze MNN-BU and find its drawbacks as follows.

1) The endmembers are hard-coded into the network via a learnable endmember matrix during training. This means, once the network is learned, the endmembers are fixed at the testing stage, possibly causing incompatibility between endmembers and abundances, especially in complex scenarios with perturbations.
2) MNN-BU only implements unrolling iterative optimization procedure of abundances and composes two sequential steps: abundance estimation and reconstructing the input with a learnable endmember matrix. This makes the network short in fully absorbing the prior knowledge embodied by the model. Therefore, the network needs more parameters and training samples to cover the practical complexity of unmixing.

In order to address the above problems, this article proposes a novel interpretable deep alternating neural network for hyperspectral unmixing. It not only equips with a more powerful physical model but also conforms to the alternating minimization of L_p-NMF. All the parameters and submodules are interpretable and can be connected with the optimization of L_p-NMF and unmixing problem. Specifically, based on LMM, we first build a nonconvex L_p $(0 < p < 1)$ sparse NMF model and derive its alternating iterative update rules following the proximal gradient algorithm. Due to the nonconvex L_p sparse inducer, the model is capable of producing more accurate abundances. Afterward, the update rules of both endmembers and abundances are unrolled to yield an alternating network (SNMF-Net), including End-Net and Abun-Net submodules, as shown in Fig. 1. SNMF-Net is in line with the alternating optimization of L_p-NMF, and no longer hard-codes the endmembers as a learnable matrix but alternatively predicts them. In this way, the network is more capable of leveraging the knowledge embodied by the model and producing more accurate endmembers and abundances at testing. Finally, all the parameters, including p, are learned by end-to-end learning, alleviating the hyperparameters selection and providing more adaptation to data.

The high interpretability of SNMF-Net makes it easy to understand its unmixing mechanism and combine the prior knowledge on the endmembers and abundances into the network for enhanced learning. Driven by recent theoretical research on the relationship between the network parameters and dictionaries in the learned LISTA [40]–[43], we further precompute and initialize the values of weight matrices using mutual coherence minimization in the sparse representation theory. This simplifies the network learning and improves the robustness of the network to noises and the effectiveness of unmixing. Enhanced physical model, advanced network architecture, and improved learning strategies help SNMF-Net dominate several alternatives methods on both synthetic data and real-world data.

This article is organized as follows. Section II introduces the L_p sparsity-based unmixing model and its alternating

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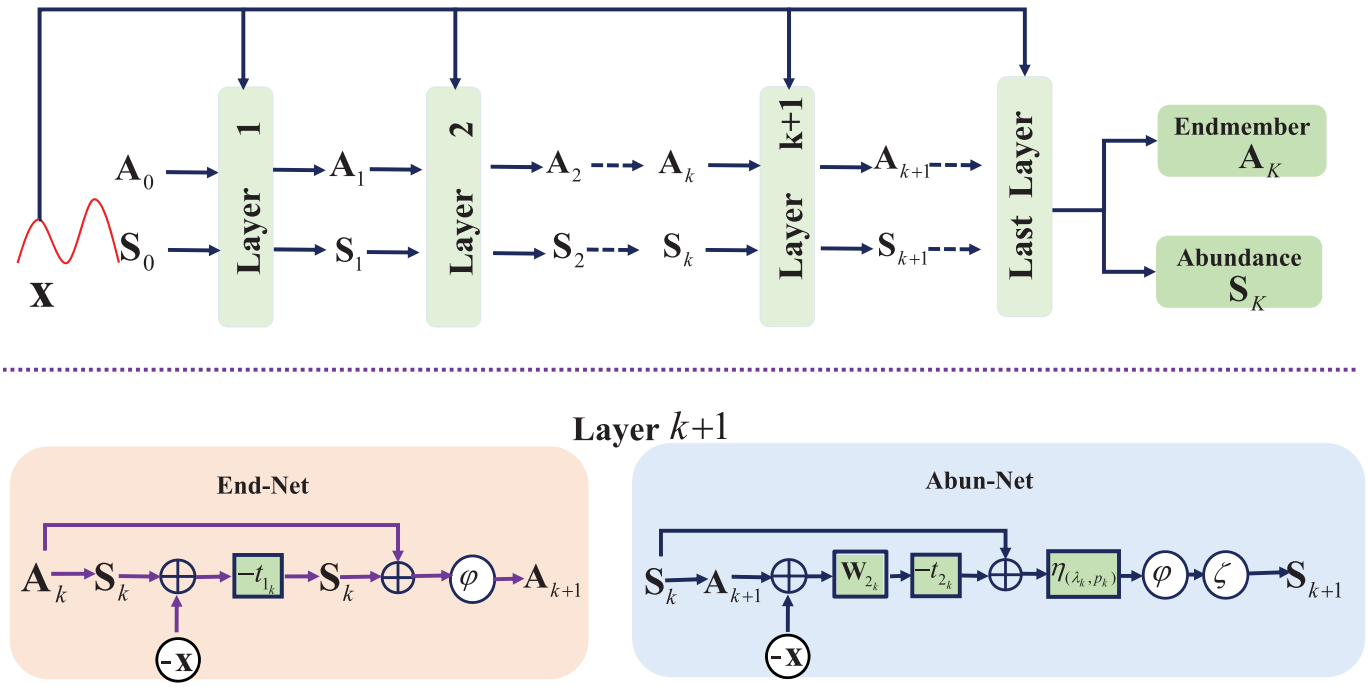


Fig. 1. Framework of the proposed SNMF-Net. Based on the L_p-norm-based sparse unmixing model, SNMF-Net takes the observed spectrum as the input and alternately predicts the endmembers and abundances using End-Net and Abun-Net submodules. All the parameters (labeled green) can be linked with the optimization of L_p-NMF and the unmixing problem and support data-driven training.

iterative optimization. In Section III, we derive the network architecture of SNMF-Net and present its learning and implementation details. Section IV reports the experimental results on both synthetic and real-world data. This article concludes in Section V.

II. L_p-NMF Unmixing Model

In this section, we introduce the L_p-NMF-based unmixing model and derive its optimization procedure.

A. Linear Spectral Mixture Model

The LMM is a widely accepted physical model describing the interaction between endmembers. The LMM assumes that the observed spectrum can be linearly represented by a set of constituent endmembers, whose weights are the fractional abundances. Given an HSI with L bands, LMM can be formally formulated as

$$\mathbf{x} = \mathbf{A}\mathbf{s} + \mathbf{e} \tag{1}$$

where $\mathbf{x} \in \mathbb{R}^{L\times 1}$ is the spectrum vector, $\mathbf{A} \in \mathbb{R}^{L\times R}$ is the endmember matrix with R materials, $\mathbf{s} \in \mathbb{R}^{R\times 1}$ contains the fractional abundances of the endmembers for $\mathbf{x}$, and $\mathbf{e} \in \mathbb{R}^{L\times 1}$ accounts for measurement errors. Using matrix notation, (1) can be reexpressed as

$$\mathbf{X} = \mathbf{A}\mathbf{S} + \mathbf{E} \tag{2}$$

where $\mathbf{X}$ is an HSI containing L bands and N pixels, $\mathbf{S} \in \mathbb{R}^{R\times N}$ is its corresponding fractional abundances, and $\mathbf{E} \in \mathbb{R}^{L\times N}$ is the additional matrix factoring in noises.

Two constraints are usually added to (2), i.e., abundance nonnegativity constraint (ANC) and abundance sum-to-one constraint (ASC) for physically meaningful unmixing. ANC means that the contribution of each endmember should be never negative, and ASC limits the entire contributions of endmembers to be equal to one. Mathematically, they are given by

$$\text{ANC}: \ \mathbf{S} \succeq 0 \quad \text{ASC}: \ \mathbf{1}_R^T\mathbf{S} = \mathbf{1}_N^T \tag{3}$$

where $\mathbf{1}$ is a vector of all ones.

B. L_p-NMF for Blind Unmixing

NMF is a prevailing tool to solve (2) by decomposing the observed HSI $\mathbf{X}$ into the product of two factor matrices, respectively, corresponding to endmembers ($\mathbf{A}$) and abundances ($\mathbf{S}$). Specifically, NMF casts hyperspectral unmixing as an optimization problem that minimizes the cost function

$$\mathcal{L}(\mathbf{A},\mathbf{S}) = \frac{1}{2}\|\mathbf{X}-\mathbf{A}\mathbf{S}\|_F^2 \ \text{s.t.} \ \mathbf{1}_R^T\mathbf{S} = \mathbf{1}_N^T, \mathbf{A}, \mathbf{S} \succeq 0. \tag{4}$$

Due to the nonconvex objective function of NMF with respect to two variables simultaneously, various priors, such as sparsity and spatial structural smoothness, are embedded into the formulation of NMF to narrow and regularize the solution to a plausible one [44]–[46]. In particular, abundance sparsity prior assumes that only a subset of endmembers contributes to a pixel. Accordingly, the objective function of NMF can be reformulated as

$$\mathcal{L}(\mathbf{A},\mathbf{S}) = \frac{1}{2}\|\mathbf{X}-\mathbf{A}\mathbf{S}\|_F^2 + \lambda f(\mathbf{S}) \ \text{ s.t. } \mathbf{1}_R^T\mathbf{S} = \mathbf{1}_N^T, \mathbf{A}, \mathbf{S} \succeq 0 \tag{5}$$

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where $f(\cdot)$ is a sparsity promoting function and λ balances the sparsity constraint term and the data fidelity term.

Generally, $f(\cdot)$ can be realized by L_0-norm of $\mathbf{S}$, which is, however, intractable because of its NP-hard nature. An alternative approach is to use L_p-norm ($0 < p \leq 1$), resulting in the following L_p-NMF-based unmixing model:

$$\mathcal{L}(\mathbf{A}, \mathbf{S}) = \frac{1}{2}\|\mathbf{X} - \mathbf{AS}\|_F^2 + \lambda\|\mathbf{S}\|_p \quad \text{s.t. } \mathbf{1}_R^T\mathbf{S} = \mathbf{1}_N^T, \mathbf{A}, \mathbf{S} \succeq 0 \tag{6}$$

where

$$\|\mathbf{S}\|_p = \sum_{r=1}^{R}\sum_{n=1}^{N}(s_{\mathrm{rn}})^p. \tag{7}$$

C. Model Optimization

Generally, many algorithms can be used to solve (6), for example, the alternating direction method of multipliers (ADMM) [47] and the alternating least-squares (ALS) algorithm [48]. In this article, we use the ALS algorithm considering its fast convergence speed and simple implementation. ALS alternately optimizes the endmembers and abundances with the other variable fixed. Specifically, fixing $\mathbf{S}$, the subproblem for optimizing $\mathbf{A}$ corresponds to

$$\mathbf{A}_{k+1} \leftarrow \min_{\mathbf{A}} \frac{1}{2}\|\mathbf{X} - \mathbf{AS}_k\|_F^2, \quad \text{s.t., } \mathbf{A} \succeq 0 \tag{8}$$

where k refers to the iteration number. This is a quadratic programming problem with a nonnegative constraint and can be iteratively solved through the gradient descent step for the quadratic part followed by a projection step for the nonnegative constraint. Consequently, the updating rule of $\mathbf{A}$ can be obtained, i.e.,

$$\mathbf{A}_{k+1} \leftarrow \left[\mathbf{A}_k - t_{1_k}(\mathbf{A}_k\mathbf{S}_k - \mathbf{X})\mathbf{S}_k^T\right]_+ \tag{9}$$

where t_{1_k} is the step size and $[\cdot]_+ = \max(\cdot, 0)$ guarantees the nonnegativity of $\mathbf{A}$.

Similarly, when $\mathbf{A}$ is fixed, we arrive at the optimization problem for $\mathbf{S}$

$$\mathbf{S}_{k+1} \leftarrow \min_{\mathbf{S}} \frac{1}{2}\|\mathbf{X} - \mathbf{A}_{k+1}\mathbf{S}\|_F^2 + \lambda\|\mathbf{S}\|_p, \quad \text{s.t., } \mathbf{S} \succeq 0, \mathbf{1}_R^T\mathbf{S} = \mathbf{1}_N^T. \tag{10}$$

Equation (10) can be decomposed into two parts, respectively, accounting for nonnegative sparsity constraint and sum-to-one constraint. The solution of $\mathbf{S}$ concerning nonnegative sparsity constraint is

$$\mathbf{S}_{k+1} \leftarrow \left[\mathrm{GST}_{(\lambda,p)}\left(\mathbf{S}_k - t_{2_k}\mathbf{A}_{k+1}^T(\mathbf{A}_{k+1}\mathbf{S}_k - \mathbf{X})\right)\right]_+ \tag{11}$$

where t_{2_k} is also the step size. Parameterized by λ and p, $\mathrm{GST}_{(\lambda,p)}(\cdot)$ is the generalized shrinkage thresholding (GST) operator for solving the following L_p-norm regularized sparse minimization problem:

$$\min_s \frac{1}{2}(y - s)^2 + \lambda|s|^p. \tag{12}$$

The close-form definition of $\mathrm{GST}_{(\lambda,p)}(\cdot)$ is

$$s = \begin{cases} \mathrm{sign}(y)S_p^{\mathrm{GST}}(|y|; \lambda), & \text{if } |y| > \Delta_p^{\mathrm{GST}}(\lambda) \\ 0, & \text{otherwise} \end{cases} \tag{13}$$

where the threshold $\Delta_p^{\mathrm{GST}}(\lambda)$ is given by

$$\Delta_p^{\mathrm{GST}}(\lambda) = (2\lambda(1-p))^{\frac{1}{2-p}} + \lambda p(2\lambda(1-p))^{\frac{p-1}{2-p}}. \tag{14}$$

Zuo *et al.* [49] have given the solution of $\mathrm{GST}(|y|; \lambda)$ by performing the following iteration steps:

$$x_{t+1} \leftarrow |y| - \lambda p x_t^{p-1} \tag{15}$$

where $x_0 = |y|$.

After every iteration, the columns of $\mathbf{S}_{k+1}$ are normalized to satisfy the ASC constrain, i.e.,

$$S_{rn} \leftarrow \frac{S_{\mathrm{rn}}}{\sum_{r=1}^{R} S_{\mathrm{rn}}}. \tag{16}$$

Equation (16) can also serve as a normalization to prevent the trivial solution that $\mathbf{S}$ is extremely small, while $\mathbf{A}$ is arbitrarily large.

Up until now, under the framework of LMM, we have built the L_p-NMF-based unmixing model and induced its corresponding iterative updating rules. Due to LMM, this model is highly interpretable, and two factor matrices can be, respectively, explained as endmembers and abundances. Generally, the endmembers and abundances can be obtained by alternately executing (9), (11), and (16) for numbers of iterations, which is always computationally demanding. Moreover, selecting proper parameters, such as t_{1_k}, t_{2_k}, λ, and p, is also troublesome but critical for this algorithm. By contrast, the DNN has a higher learning capacity, and the feedforward process is very computationally efficient because no iteration is needed. To this end, we construct a neural network by unrolling the iterative updating rules in (9), (11), and (16) while retaining the attractive interpretation of the L_p-NMF-based unmixing model.

III. SNMF-Net

In this section, we first build the connection between the proximal gradient method and DNN, then introduce the details of SNMF-Net, and address its learning and implementation issues.

A. Connection Between Proximal Gradient Method and DNN

As shown in Fig. 1, the proposed SNMF-Net is a joint LMM-based network that alternately optimizes endmembers and abundances by End-Net and Abun-Net. Both End-Net and Abun-Net are obtained by unrolling the iterative optimization of $\mathbf{A}$ and $\mathbf{S}$. The optimization problems of $\mathbf{A}$ and $\mathbf{S}$ both in essence share the following unconstrained problem:

$$\min_{\mathbf{u}} \frac{1}{2}\|\mathbf{Bu} - \mathbf{v}\|_2^2 + f(\mathbf{u}) \tag{17}$$

where the first part is convex and differentiable, representing the reconstruction term, and the second part is neither necessarily convex nor differentiable but has a fast proximal operator, representing the regularization term.

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The proximal gradient algorithm is a widely used framework for such problem with the following update rule comprising of two fundamental steps, i.e.:

$$\mathbf{u}' \leftarrow \mathbf{u}_k - t\mathbf{B}^T(\mathbf{B}\mathbf{u}_k - \mathbf{v}) \tag{18}$$

$$\mathbf{u}_{k+1} \leftarrow \mathbf{prox}_{tf}(\mathbf{u}') \tag{19}$$

where **prox** is the proximal operator, for example, soft-thresholding operator, and $t > 0$ is the step size. As pointed out by Xie *et al.* [50], the upper bound of t is $(1/\|\mathbf{B}^T\mathbf{B}\|_2)$ to satisfy the strict convergence of the proximal gradient method. Running (18) and (19), a number of iterations till convergence yields the solution of $\mathbf{u}$.

From another point of view, the two steps of updating $\mathbf{u}$ can be linked with a typical local block of a recurrent neural network (RNN). Concretely, (18) accounts for a linear operation to compute an update in the direction of the gradient, and (19) mimics an elementwise nonlinearity with a proximal operator **prox**. Taking the soft-thresholding operator as an example, it can be defined as $\mathcal{S}_\lambda(u) = \mathrm{ReLu}(x - \lambda) - \mathrm{ReLu}(-x - \lambda)$. When the number of iterations truncated to K and unrolling the RNN for K times, the network becomes a K-layer feedforward neural network and supports discriminative learning. Introducing two differentiable variables, respectively, responsible for step size t and $\mathbf{W} = \mathbf{B}^T$ as network parameters, (18) and (19) can be unrolled into a network with consecutive layers being a function parameterized by $\{\mathbf{W}, t\}$ such that

$$\mathbf{u}_{k+1} = \varphi(\mathbf{u}_k - t\mathbf{W}(\mathbf{B}\mathbf{u}_k - \mathbf{v})). \tag{20}$$

Here, $\varphi(\cdot)$ is a nonlinear activation function, and k indexes the layer corresponding to the kth iteration in the iterative update procedure.

B. L_p-NMF Inspired Deep Alternating Neural Network

Following the above perspective, we introduce the detailed network architecture for endmembers and abundances.

1) End-Net for Endmembers: Based on the iterative update in (9), the unrolled subnetwork for endmember can be defined by recursion

$$\mathbf{A}_{k+1} = \varphi\left(\mathbf{A}_k - t_{1_k}\left(\mathbf{A}_k\mathbf{S}_k - \mathbf{X}\right)\mathbf{W}_{1_k}\right) \tag{21}$$

where $\varphi(\cdot) = \max(\cdot)$ is the ReLu activation function allowing for the nonnegativity of $\mathbf{A}$.

From (21), it can be found that $\mathbf{W}_{1_k}$ shares the same size with the input samples, indicating that the network can only process a fixed number of pixels at a time in order to satisfy matrix multiplication. In practice, we always expect that the network can accept any number of pixels for testing. Moreover, End-Net produces a set of endmembers given a set of pixels, and the endmembers may change when provided with a different set of pixels. This is unreasonable for the unmixing task because all the pixels in a scene should share the same set of endmembers. To satisfy the above requirement, we instead replace $\mathbf{W}_{1_k}$ with $\mathbf{S}_k^T$ so that the network is able to accept any number of pixels, and all the pixels share the same set of endmembers. Accordingly, the network defined in (21) follows the iteration

$$\mathbf{A}_{k+1} = \varphi\left(\mathbf{A}_k - t_{1_k}(\mathbf{A}_k\mathbf{S}_k - \mathbf{X})\mathbf{S}_k^T\right). \tag{22}$$

Unlike the original network defined in (21), only the step sizes $\{t_{1_k}\}$ are required to learn in the network defined by (22), significantly simplifying network training.

2) Abun-Net for Abundances: Different from $\mathbf{A}$, the update rule of $\mathbf{S}$ contains an additional iterative optimization of S_p^{GST}, which makes it very difficult to analyze the relationship between $\mathbf{S}_k$ and (λ, p). For simplicity, we only iterate (15) for one time, yielding the following solution for GST:

$$\eta_{(\lambda,p)}(y) = \begin{cases} \mathrm{sign}(y)(|y| - \lambda p(|y|)^{p-1}), & \text{if } |y| > \Delta_p^{\mathrm{GST}}(\lambda) \\ 0, & \text{otherwise.} \end{cases} \tag{23}$$

Setting different parameters per iteration generally helps yield more favorable performance and accelerate the convergence. This strategy has been commonly employed, especially for nonconvex problems [16], [51]–[53]. Moreover, decouple learning across layers also is beneficial to overcome gradient explosion and vanishing problems. To this end, we adopt this strategy and choose different parameters $\{\lambda_k, p_k\}$ across the iteration of $\mathbf{S}$. Consequently, the updating rule for $\mathbf{S}_{k+1}$ in (11) becomes

$$\mathbf{S}_{k+1} \leftarrow \left[\eta_{(\lambda_k,p_k)}\left(\mathbf{S}_k - t_{2_k}\mathbf{A}_{k+1}^T(\mathbf{A}_{k+1}\mathbf{S}_k - \mathbf{X})\right)\right]_+. \tag{24}$$

Accordingly, the succeeding layers in Abun-Net are given by

$$\mathbf{S}_{k+1} = \zeta\left(\varphi\left(\eta_{(\lambda_k,p_k)}\left(\mathbf{S}_k - t_{2_k}\mathbf{W}_{2_k}(\mathbf{A}_{k+1}\mathbf{S}_k - \mathbf{X})\right)\right)\right) \tag{25}$$

where $\zeta(\cdot)$ is a normalization layer satisfying ASC in (16). The parameter set of Abun-Net is $\{t_{2_k}, \mathbf{W}_{2_k}, p_k, \lambda_k\}$. Instead of inefficiently tuning p to obtain the desired sparsity level, p is also layerwise learned from data, offering better adaptation to the underlying properties of the data.

In summary, based on the LMM model, an L_p-NMF unmixing model is first built. Inspired by the connection between the PG optimization method and DNN, we unroll the optimization of L_p-NMF for $\mathbf{A}$ and $\mathbf{S}$ to yield a K-layer alternating network dubbed SNMF-Net for blind unmixing. Different from alternative deep learning-based unmixing methods, SNMF-Net is physically interpretable. On the one hand, SNMF-Net is under the framework of LMM, which considers the light interaction between materials and carefully designed optimization. On the other hand, all the key parameters can be seamlessly linked with the associate optimization and support end-to-end learning. Due to such interpretation, all the prior knowledge about the unmixing task, optimization algorithms, and sparse representation theory can be easily integrated into the network for robust learning.

C. Network Training

In blind unmixing, both $\mathbf{A}$ and $\mathbf{S}$ are unknown and, thereby, it's unrealistic to directly learn them by discriminative training. Instead, we take the Euclidean distance between the reconstructed spectrum and input spectrum as the loss function. Motivated by the phenomenon that adding auxiliary classifiers to intermediate layers helps to propagate gradients back through all the layers [54], we take the output of all

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the layers into the loss function. Given the training dataset $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$, including N spectrum, the loss function can be defined as

$$\mathcal{L}_\Theta = \frac{1}{2N}\sum_{k=1}^{K}\|\mathbf{X} - \mathbf{A}_k\mathbf{S}_k\|_F^2 \tag{26}$$

where $\Theta = \{t_{1_k}, t_{2_k}, \mathbf{W}_{2_k}, p_k, \lambda_k\}$ are the parameters to be learned and K is the number of layers. To learn the network, the partial derivative of $\mathcal{L}_\Theta$ concerning each variable should be computed first. For simplicity, we consider an individual training sample $\mathbf{x}$ and define the following intermediate variables:

$$\begin{aligned}
\widetilde{\mathbf{A}}_k &= \mathbf{A}_{k-1} - t_{1_{k-1}}(\mathbf{A}_{k-1}\mathbf{s}_{k-1} - \mathbf{x})\mathbf{s}_{k-1}^T \\
\widetilde{\mathbf{s}}_k &= \mathbf{s}_{k-1} - t_{2_{k-1}}\mathbf{W}_{2_{k-1}}(\mathbf{A}_k\mathbf{s}_{k-1} - \mathbf{x}) \\
\widehat{\mathbf{s}}_k &= \varphi\big(\eta_{(\lambda_{k-1},p_{k-1})}(\widetilde{\mathbf{s}}_k)\big).
\end{aligned} \tag{27}$$

The partial differentials of $\mathcal{L}_\Theta$ with respect to each variable are given as follows:

$$\begin{aligned}
&\frac{\partial\mathcal{L}}{\partial t_{2_k}} = \frac{\partial\widetilde{\mathbf{s}}_{k+1}}{\partial t_{2_k}}\frac{\partial\mathcal{L}}{\partial\widetilde{\mathbf{s}}_{k+1}} \quad \frac{\partial\mathcal{L}}{\partial p_k} = \frac{\partial\widetilde{\mathbf{s}}_{k+1}}{\partial p_k}\frac{\partial\mathcal{L}}{\partial\widetilde{\mathbf{s}}_{k+1}} \\
&\frac{\partial\mathcal{L}}{\partial\lambda_k} = \frac{\partial\widetilde{\mathbf{s}}_{k+1}}{\partial\lambda_k}\frac{\partial\mathcal{L}}{\partial\widetilde{\mathbf{s}}_{k+1}} \quad \frac{\partial\mathcal{L}}{\partial\mathbf{W}_{2_k}} = \frac{\partial\mathcal{L}}{\partial\widetilde{\mathbf{s}}_{k+1}}\frac{\partial\widetilde{\mathbf{s}}_{k+1}}{\partial\mathbf{W}_{2_k}} \\
&\frac{\partial\mathcal{L}}{\partial\widetilde{\mathbf{s}}_{k+1}} = \frac{\partial\widehat{\mathbf{s}}_{k+1}}{\partial\widetilde{\mathbf{s}}_{k+1}}\frac{\partial\mathbf{s}_{k+1}}{\partial\widehat{\mathbf{s}}_{k+1}}\frac{\partial\mathcal{L}}{\partial\mathbf{s}_{k+1}} \quad \frac{\partial\mathcal{L}}{\partial\widetilde{\mathbf{A}}_{k+1}} = \frac{\partial\mathcal{L}}{\partial\mathbf{A}_{k+1}}\frac{\partial\mathbf{A}_{k+1}}{\partial\widetilde{\mathbf{A}}_{k+1}} \\
&\frac{\partial\mathcal{L}}{\partial t_{1_k}} = \frac{\partial\widetilde{\mathbf{A}}_{k+1}}{\partial t_{1_k}}\frac{\partial\mathcal{L}}{\partial\widetilde{\mathbf{A}}_{k+1}}.
\end{aligned} \tag{28}$$

$(\partial\mathbf{A}_{k+1}/\partial\widetilde{\mathbf{A}}_{k+1})$ and $(\partial\widehat{\mathbf{s}}_{k+1}/\partial\widetilde{\mathbf{s}}_{k+1})$ are both related to corresponding activation functions and can be, respectively, obtained by

$$\frac{\partial\mathbf{A}_{k+1}}{\partial\widetilde{\mathbf{A}}_{k+1}} = \mathcal{I}_{\widetilde{\mathbf{A}}_{k+1}>0} \qquad \frac{\partial\widehat{\mathbf{s}}_{k+1}}{\partial\widetilde{\mathbf{s}}_{k+1}} = \mathcal{I}_{\eta_{(\lambda_k,p_k)}(\widetilde{\mathbf{s}}_{k+1})>0} \tag{29}$$

where $\mathcal{I}$ is an indicator function. According to (27), we can get

$$\begin{aligned}
\frac{\partial\widetilde{\mathbf{s}}_{k+1}}{\partial t_{2_k}} &= \mathbf{W}_{2_k}(\mathbf{x} - \mathbf{A}_{k+1}\mathbf{s}_{k+1}) \\
\frac{\partial\widetilde{\mathbf{A}}_{k+1}}{\partial t_{1_k}} &= (\mathbf{x} - \mathbf{A}_k\mathbf{s}_k)\mathbf{s}_k^T \\
\frac{\partial\widetilde{\mathbf{s}}_{k+1}}{\partial\mathbf{W}_{2_k}} &= t_{2_k}(\mathbf{x} - \mathbf{A}_{k+1}\mathbf{s}_k).
\end{aligned} \tag{30}$$

Based on (23), each element of $(\partial\widetilde{\mathbf{S}}_{k+1}/\partial p_{k+1})$ and $(\partial\widetilde{\mathbf{S}}_{k+1}/\partial\lambda_{k+1})$ can, respectively, be derived by

$$\frac{\partial\widetilde{s}_{k+1}}{\partial p_k} = \begin{cases} 0, & \widetilde{s}_{k+1} \le \Delta_{p_k}^{\mathrm{GST}}(\lambda_k) \\ -\lambda\mathrm{sign}(\widetilde{s}_{k+1})|\widetilde{s}_{k+1}|^{p-1} & \\ (p\ln(|\widetilde{s}_{k+1}|)+1), & \text{otherwise} \end{cases} \tag{31}$$

$$\frac{\partial\widetilde{s}_{k+1}}{\partial\lambda_k} = \begin{cases} 0, & \widetilde{s}_{k+1} \le \Delta_{p_k}^{\mathrm{GST}}(\lambda_k) \\ -\mathrm{sign}(\widetilde{s}_{k+1})p(|\widetilde{s}_{k+1}|)^{p-1} & \text{otherwise.} \end{cases} \tag{32}$$

Based on (16), $(\partial\mathbf{s}_{k+1}/\partial\widehat{\mathbf{s}}_{k+1})$ is

$$\frac{\partial\mathbf{s}_{k+1}}{\partial\widehat{\mathbf{s}}_{k+1}} = \frac{1}{T}\mathbf{I} - \frac{1}{T^2}\widehat{\mathbf{s}}_{k+1}\mathbf{1}_R^T \tag{33}$$

where $T = \sum\widehat{\mathbf{s}}_{k+1}$.

$(\partial\mathcal{L}/\partial\mathbf{A}_k)$ and $\frac{\partial\mathcal{L}}{\partial\mathbf{s}_k}$ can be, respectively, computed as

$$\begin{aligned}
\frac{\partial\mathcal{L}}{\partial\mathbf{s}_K} &= \mathbf{A}_K^T(\mathbf{A}_K\mathbf{s}_K - \mathbf{x}) \\
\frac{\partial\mathcal{L}}{\partial\mathbf{A}_K} &= (\mathbf{A}_K\mathbf{s}_K - \mathbf{x})\mathbf{s}_K^T - t_{2_{K-1}}\mathbf{W}_{2_{K-1}}^T\frac{\partial\mathcal{L}}{\partial\widetilde{\mathbf{s}}_K}\mathbf{s}_{K-1}^T \\
\frac{\partial\mathcal{L}}{\partial\mathbf{s}_k} &= \mathbf{A}_k^T(\mathbf{A}_k\mathbf{s}_k - \mathbf{x}) + t_{1_k}(\mathbf{x} - 2\mathbf{A}_k\mathbf{s}_k)\frac{\partial\mathcal{L}}{\partial\widetilde{\mathbf{A}}_{k+1}} \\
&\quad + \big(\mathbf{I} - t_{2_k}\mathbf{W}_{2_k}\mathbf{A}_{k+1}\big)\frac{\partial\mathcal{L}}{\partial\widetilde{\mathbf{s}}_{k+1}}, \quad k = 1:K-1 \\
\frac{\partial\mathcal{L}}{\partial\mathbf{A}_k} &= (\mathbf{A}_k\mathbf{s}_k - \mathbf{x})\mathbf{s}_k^T + \frac{\partial\mathcal{L}}{\partial\widetilde{\mathbf{A}}_{k+1}}\big(\mathbf{I} - t_{1_k}\mathbf{s}_k\mathbf{s}_k^T\big) \\
&\quad - t_{2_{k-1}}\mathbf{W}_{2_{k-1}}^T\frac{\partial\mathcal{L}}{\partial\widetilde{\mathbf{s}}_k}\mathbf{s}_{k-1}^T, \quad k = 1:K-1.
\end{aligned} \tag{34}$$

Substituting (29)–(32) and (34) into (28), we can get the derivations of all the variables. Once they are obtained, all the parameters in Θ can be learned by end-to-end training via any gradient descent method. The training process can be summarized as follows.

1) Estimate the endmember matrix $\mathbf{A}_0$ using any endmember extraction algorithm, e.g., VCA.
2) Randomly sample a number of pixels to construct the training set $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$, and calculate their initial abundances $\mathbf{S}_0$ by any traditional unmixing methods, for example, FCLS.
3) Initialize the network with $\mathbf{A}_0$ and $\mathbf{S}_0$. In contrast to alternative deep learning-based methods, the embedded models make it easy to integrate the prior knowledge on unmixing into the network for enhanced learning.
4) Train SNMF-Net with stochastic gradient descent algorithm till convergence to learn the network parameters.

D. Network Implementation

It is always expected to incorporate the additional auxiliary knowledge at hand to guide the network learning especially when the training samples are small. Due to the high interpretability of SNMF-Net, prior knowledge can be reasonably integrated into the network to achieve satisfactory unmixing. NMF-based unmixing methods usually initialize the factor matrices with the endmembers extracted by VCA [6] and abundance estimated by FCLS [7]. For this reason, we also adopt this strategy to initialize $\mathbf{A}_0$ and $\mathbf{S}_0$. After that, t_{1_k} and t_{2_k} are accordingly set as $(1/\|\mathbf{S}_0\mathbf{S}_0^T\|_2 + \epsilon)$ and $(1/\|\mathbf{A}_0^T\mathbf{A}_0\|_2 + \epsilon)$, where ϵ is a tiny number given by 0.01. p is initially assigned as 0.5 based on the observation in [12].

We follow the mutual coherence minimization in the sparse representation theory to initialize $\mathbf{W}_{2_k}$. In sparse representation, mutual coherence is an important indicator to depict the sparse recovery quality of a dictionary defined as

$$\mu(\mathbf{D}) = \max_{1\le i\ne j\le L}\frac{\mathbf{D}_i^T\mathbf{D}_j}{\|\mathbf{D}_i\|_2\|\mathbf{D}_j\|_2}. \tag{35}$$

A dictionary with low coherence implies better sparse recovery ability. Based on the above definition, we define a generalized mutual coherence (GMC) as follows.

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Definition 1: For a dictionary $\mathbf{D} \in \mathbb{R}^{L\times R}$, its GMC is given by

$$\hat{\mu}(\mathbf{D}) = \inf_{\substack{\mathbf{W}\in\mathbb{R}^{L\times R}\\ \mathbf{W}_i^T\mathbf{D}_i=1, 1\le i\le R}} \left\{ \max_{1\le i\ne j\le R} \mathbf{W}_i^T\mathbf{D}_j \right\}. \tag{36}$$

As pointed out in [42], if $\mathbf{W} \in \mathcal{W}(\mathbf{D})$, where $\mathcal{W}(\mathbf{D}) = \{\mathbf{W} \in \mathbb{R}^{L\times R} : \textit{attains the infimum in (36)}\}$, the sparsity recovery error is bounded, and $\mathbf{W}$ can be precomputed by solving the following problem:

$$\arg\min_{\mathbf{W}\in\mathbb{R}^{L\times R}} \|\mathbf{W}^T\mathbf{D}\|_F^2 \quad \text{s.t.}\mathbf{W}^\mathbf{T}\mathbf{D} = \mathbf{I}. \tag{37}$$

In this way, there is no need to learn $\mathbf{W}$ in a data-drive manner, significantly reducing the training complexity and improving the robustness of a network.

When it terms to unmixing, the endmembers can be considered as dictionaries. Intuitively, we can compute $\mathbf{W}_{2_k}$ using (37) when the endmembers are ready. However, the endmembers are also unknown in the blind unmixing task, meaning that it is unrealistic to obtain an exact solution of $\mathbf{W}_{2_k}$ in advance. Fortunately, as mentioned above, VCA can provide a good initial estimation of endmembers. For this regard, we alternatively still keep $\mathbf{W}_{2_k}$ learnable and initial them as the values computed by (37). By this means, the prior knowledge on the spare representation theory is integrated into the network, which enhances the network learning as will be shown in Section IV.

IV. Experiments

In this section, we comprehensively evaluate the unmixing ability of the proposed SNMF-Net on both synthetic data and real-world data. Seven methods are selected as alternative methods to be compared with our method, including Arctan-NMF [17], TV-RSNMF [55], MV-NTF [18], uDAS [26], UnDIP [30], DAEN [22]. and MNN-BU [39]. Among these compared methods, Arctan-NMF, TV-RSNMF, and MV-NTF are three model-based methods. Arctan-NMF and TV-RSNMF are based on NMF. MV-NTF is a tensor-based method. uDAS and DAEN are learning-based methods based on AE. MNN-BU is a model-based method whose network architecture is obtained by unrolling the optimization of abundances. UnDIP takes advantage of deep prior knowledge embodied by DNNs to improve unmixing. All relevant parameters are set as suggested in the original implementation.

A. Performance Evaluation Criteria

The SAD and the root-mean-square error (RMSE) were used for quantitative performance evaluation. SAD measures the spectral angle between reference endmembers $\mathbf{A}_r$ and estimated endmembers $\widehat{\mathbf{A}}_r$ defined as

$$\text{SAD}_r = \arccos\left(\frac{\mathbf{A}_r^T\widehat{\mathbf{A}}_r}{\|\mathbf{A}_r\|\|\widehat{\mathbf{A}}_r\|}\right). \tag{38}$$

RMSE computes the error between the reference abundance map $\mathbf{S}_r$ and estimated map $\widehat{\mathbf{S}}_r$, mathematically formulated as

$$\text{RMSE}_r = \left(\frac{1}{N}|\mathbf{S}_r - \widehat{\mathbf{S}}_r|^2\right)^{\frac{1}{2}}. \tag{39}$$

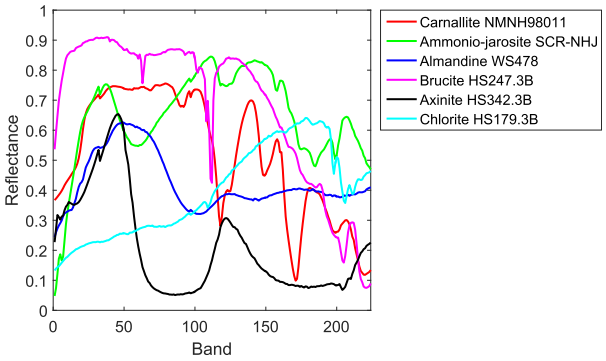


Fig. 2. Spectral signatures of six endmembers used in the synthetic data.

The small values of SAD and RMSE imply better unmixing.

B. Experiments on Synthetic Data

Six pure signatures (Carnallite, Ammonio-jarosite, Almandine, Brucite, Axinite, and Chlonte) covering 224 bands with wavelengths between 0.38 and 2.5 μm were first selected from the USGS library[1] to generate endmembers. Their spectral curves are displayed in Fig. 2. Following [19], the abundances were generated as follows.

1) A synthetic image containing $z^2 \times z^2$ pixels is segmented into z^2 nonoverlapped regions with the same size of $z \times z$.
2) Each pixel of a region is filled with two randomly selected endmembers whose ratios are, respectively, given as β and $1-\beta$.
3) A spatial low-pass Gaussian filter whose variance is set to 2 is used to generate mixed pixels, resulting in a relatively higher degree of mixing.
4) Finally, the fractions of all endmembers in each pixel are rescaled to meet the ASC constraint.

In our experiment, we set $z = 8$ and $\beta = 0.8$ to generate the synthetic data. The generated clean HSI is further degraded by zero-mean additive Gaussian to produce noisy HSI with signal-to-noise ratios (SNRs) defined as

$$\text{SNR} = 10\log_{10}\frac{E\left[\mathbf{y}^T\mathbf{y}\right]}{E\left[\mathbf{e}^T\mathbf{e}\right]} \tag{40}$$

where $\mathbf{y}$ and $\mathbf{e}$ are the clean signal and the noise at a pixel and $E[\cdot]$ denotes the expectation operator.

1) Influence of the Number of Layers: We first analyze the influence of layers on unmixing accuracy. The clean HSI was degraded by the Gaussian noise with SNR = 30 dB, and the training size was set to 500. By altering K from 3 to 30 with an interval of 3, Fig. 3 presents the changes of SAD and RMSE. It can be seen that both SAD and RMSE decrease rapidly when K increases from 3 to 9, implying the positive effect of deep models. After reaching the bottom at $K = 9$, SAD gradually increases, and RMSE almost stays steady. This may because the network parameters increase as the number of layers increases, which requires more data to learn. Since the best values of SAD and RMSE are obtained when $K = 9$,

[1]https://www.usgs.gov/labs/spec-lab/capabilities/spectral-library

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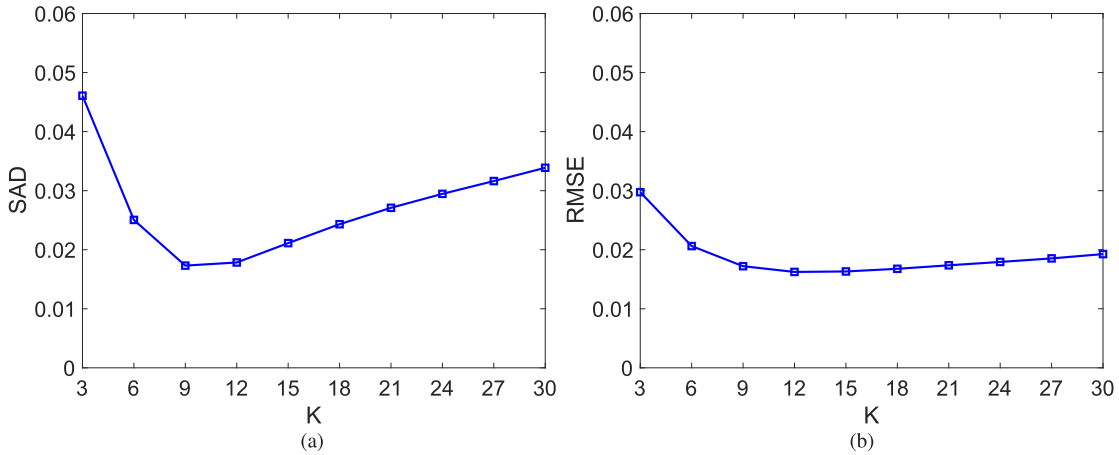


Fig. 3. (a) SAD and (b) RMSE with respect to the number of layers.

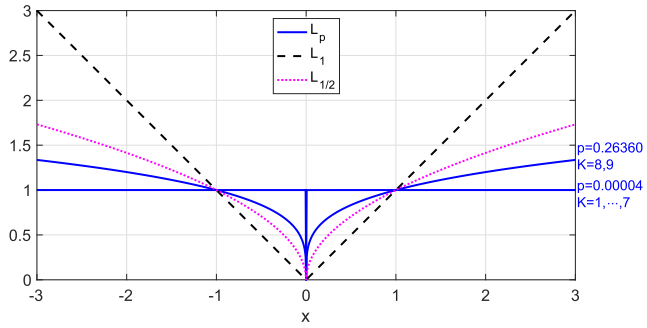


Fig. 4. Learned L_p sparse penalty with respect to K values.

we set the number of layers to 9 in the following experiments. This experiment shows that very few layers of SNMF-Net are able to produce promising unmixing, confirming its strong representing ability.

When $K = 9$, we demonstrate the learned p concerning the number of layers. As shown in Fig. 4, in the first few layers, L_p-norm is close to L_0-norm, indicating that the network tends to obtain a very sparse estimation of abundance maps, which helps to suppress the side effect of perturbation. When K is over 7, the p value increases to 0.2636, meaning that the network is learning to produce a less sparse solution. This is also in line with that, as the number of iterations increases, the optimization algorithm gradually converges, and smaller penalties should be added. Moreover, compared with the fixed L_1-norm, our network is more adaptive to the data as p is learned from the data. Unlike L_1-norm, the proposed is able to yield sparser and unbiased abundances because the gradient of learned L_p-norm decreases as x increases, which means that large magnitudes components are less possible to be shrunk to 0. In short, this experiment verifies the interpretability of the proposed SNMF-Net.

2) Influence of the Number of Training Samples: In this experiment, we investigate the impact of the number of training samples on unmixing. We set SNR $=$ 30 dB to generate the noisy HSI. Fig. 5 plots the SAD and RMSE changes with the number of training samples in the range of $\{64, 256, 500, 1000, 2000, 4096\}$. As can be seen from Fig. 5, without using all the pixels for training, i.e., only 500 pixels (12.21% of all the pixels), the network can achieve as satisfying a performance as unmixing with all pixels. This phenomenon evidently shows the strong learning and generalization ability of SNMF-Net. The main reason is that SNMF-Net takes the spectral mixture model and the alternating optimization of L_p-NMF into consideration to build the network architecture. All the key components of the proposed SNMF-Net are explainable, facilitating the integration of prior knowledge of unmixing into the network for enhanced learning.

3) Influence of Initialization: As mentioned in the network implementation, we adopt (37) to initialize $\mathbf{W}_{2_k}$, which is induced by the GMC theory to enhance learning. Here, we conducted a study on the effect of such initialization. Fixing the number of training samples as 500 and changing SNR from 15, 20, 25, and 30 dB to ∞ (noise-free), Fig. 6 reports the SAD and RMSE changes with or without GMC-based initialization. Here, "SNMF-Net without GMC" refers to setting $\mathbf{W}_{2_k} = \mathbf{A}^T$. It can be clearly observed that GMC-based initialization helps SNMF-Net achieve significantly better and robuster unmixing performance. This experiment shows the effectiveness of GMC-based initialization and the benefit of interpretable network architecture.

Attributing to the embedded physical model, the prior knowledge learned or based on domain knowledge concerning unmixing can be flexibly incorporated into the network to enhance network learning. We further initialized the network by NMF whose results are denoted as SNMF-Net-NMF. Fig. 7 compares SAD and RMSE with respect to different initializations. As can be seen, using NMF to initialize the network yields lower SAD and RMSE because NMF can provide more accurate endmembers and abundances, specifically in lower SNR cases. This experiment evidently shows the benefits of the model-based neural network, i.e., easy incorporation of prior knowledge.

4) Robustness to Noise: We further compared SNMF-Net with alternative methods to demonstrate its superiority in unmixing. The SNRs were selected as 15, 20, 25, and 30 dB to ∞ (noise-free). Based on the above experiment observations, we used 500 pixels to train SNMF-Net. The other compared

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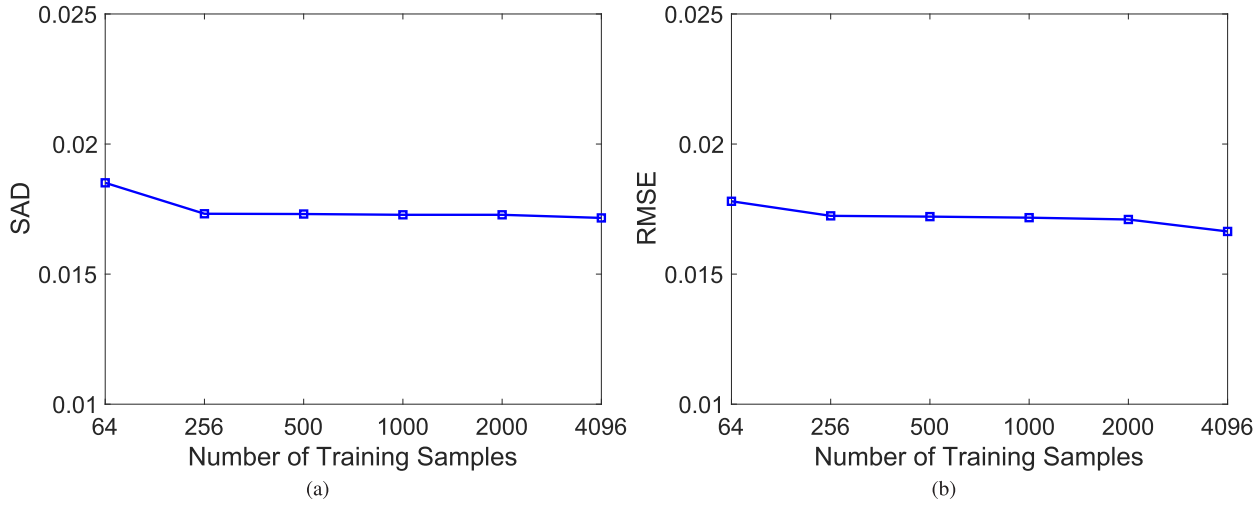


Fig. 5. (a) SAD and (b) RMSE with respect to the number of training samples.

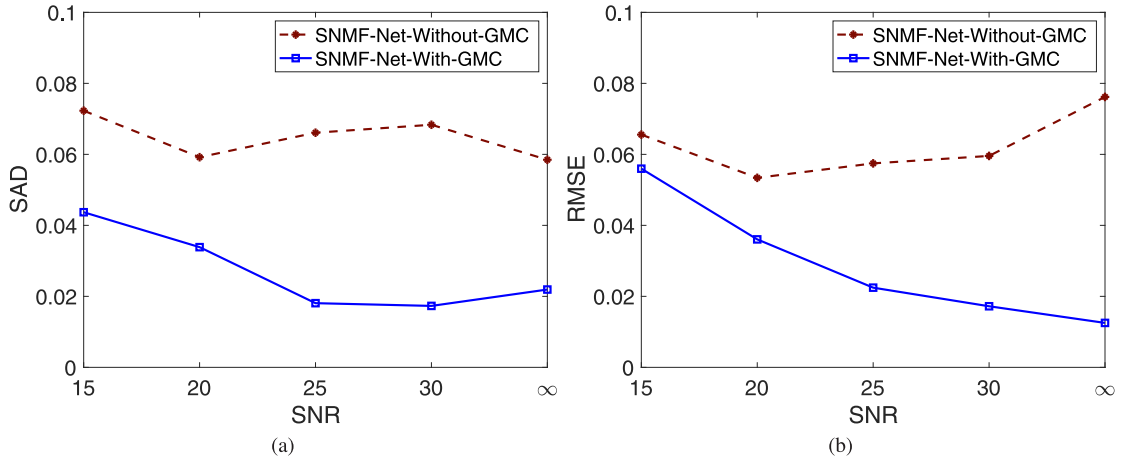


Fig. 6. Impact of GMC-based initialization. (a) SAD. (b) RMSE.

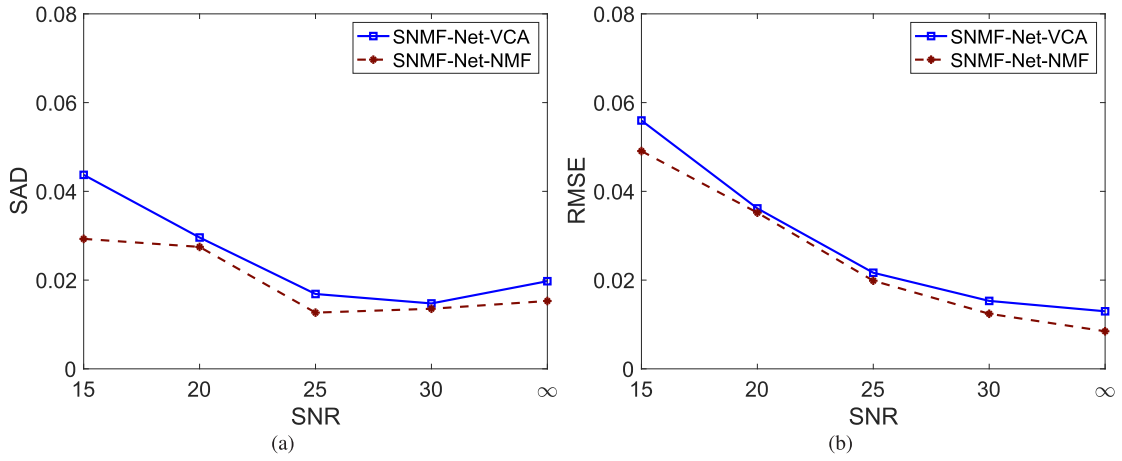


Fig. 7. Impact of different initializations. (a) SAD. (b) RMSE.

methods utilized all the pixels to conduct unmixing. Fig. 8 presents the unmixing performance of all the methods regarding SNR settings. In general, most of the unmixing methods show descending trend as the SNR increases. UnDIP provides inferior SAD and RMSE because it estimates endmembers and abundances in sequence rather than simultaneously. The failure of endmember extraction unavoidably causes poor abundance estimation. Moreover, it can be seen from the

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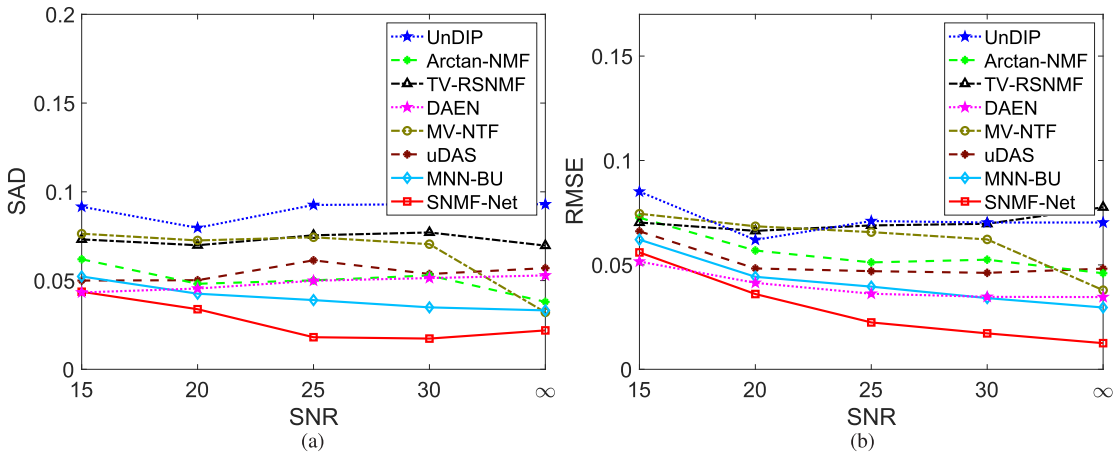


Fig. 8. Performance of all eight unmixing methods with respect to different noise levels in terms of (a) SAD and (b) RMSE.



Fig. 9. Three real-world datasets shown in gray. (a) Third band image of Jasper Ridge. (b) 80th band image of Urban. (c) Third band image of Washington DC Mall.

TABLE I
MEANS AND STANDARD DEVIATIONS OF THE SAD ON THE JASPER RIDGE DATASET

Algorithm	TV-RSNMF [55]	MV-NTF [18]	DAEN [22]	uDAS [26]	MNN-BU [39]	UnDIP [30]	SNMF-Net
Tree	0.0776±4.91%	0.2209±2.17%	0.1774±0.82%	0.1511±1.37%	**0.0468±0.32%**	0.1493±0%	0.0478 ±1.62%
Water	0.1337±1.31%	0.2352±1.65%	0.3237±1.42%	0.1742±11.39%	0.1179±0.23%	0.2529±0%	**0.0855±1.77%**
Soil	0.1047±10.11%	0.1752±4.67%	0.1123±2.03%	0.1639±6.09%	**0.0357±0.43%**	0.1144±0%	0.0443±3.09%
Road	0.6515±27.46%	0.1741±4.35%	**0.0588±1.59%**	0.0597±0.27%	0.0901±0.37%	0.2757±0%	0.0875±0.05%
Mean	0.2419±3.06%	0.2015±1.77%	0.1680±0.43%	0.1373±3.27%	0.0726±0.11%	0.1981±0%	**0.0663±1.04%**

figure that MV-NTF outperforms TV-RSNMF, especially in high SNR scenarios due to its inherent advantages of preserving the spatial–spectral structure of HSIs. Compared with three model-based methods, DAEN, uDAS, and SNMF-Net provide more desirable unmixing. Among them, the most promising unmixing results belong to the proposed SNMF-Net even using only 12.21% of pixels. This attributes to that SNMF-Net is equipped with the hybrid advantages of model-based methods and learning-based methods.

C. Experiments on Real-World Data

We further carried out experiments on three widely used real-world HSIs, Jasper Ridge, Urban, and Washington DC Mall to more comprehensively show the advantages of SNMF-Net. The same as the synthetic experiment, 500 randomly sampled pixels were adopted for training SNMF-Net. Because of the unavailability of ground-truth endmembers and abundances, we instead selected several pixels that are most likely to be pure pixels and set their spectral signatures as reference endmembers. After that, the reference abundances maps were obtained using CVX MATLAB software.

1) Jasper Ridge Dataset: It was collected by AVIRIS sensor and contains 512 × 613 pixels and 224 spectral channels, covering wavelength from 380 to 2500 nm. We considered a part of it with 100 × 100 pixels doing experiments. After removing bands corrupted by noise and low SNR and water-vapor absorption, 198 bands were retained for the experiment. Fig. 9(a) presents the third band image of Jasper Ridge HSI. In the experiment, we set four kinds of materials as endmembers, including tree, water, soil, and road.

Table I provides a quantitative comparison of all the competing methods with respect to the SAD index. From Table I,

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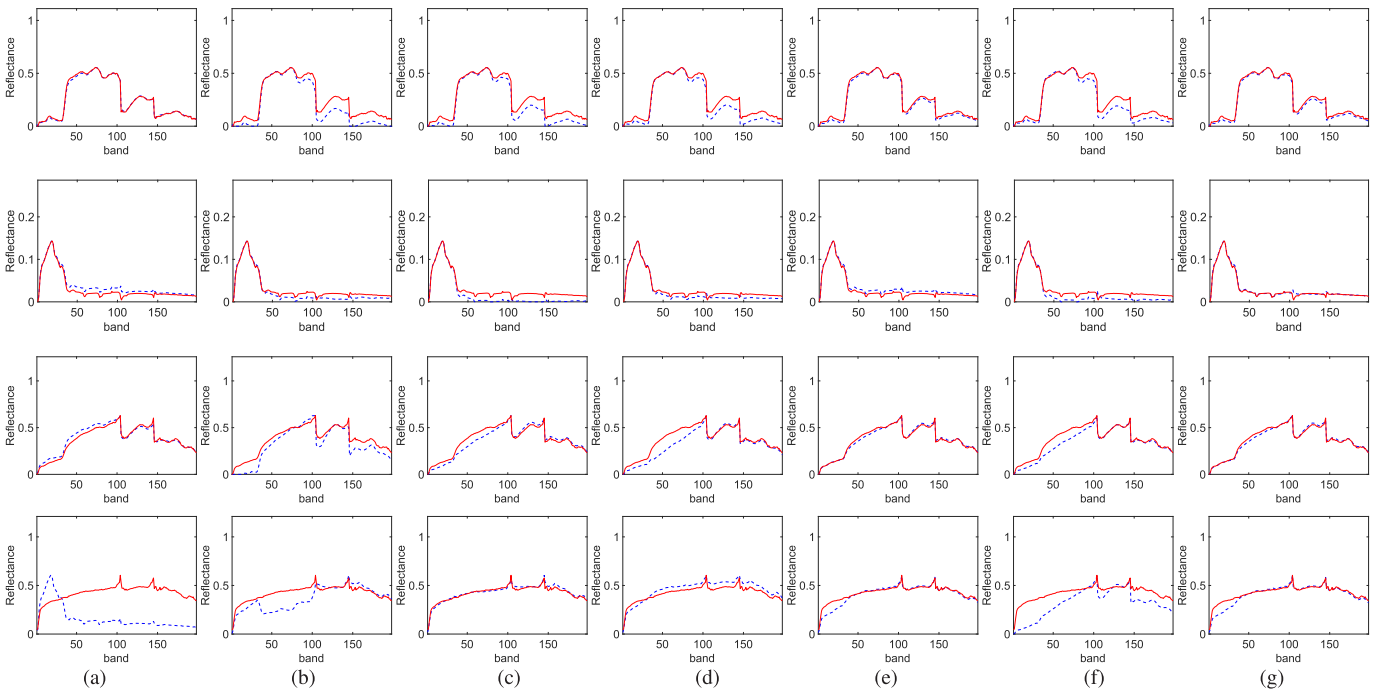


Fig. 10. Estimated endmembers by seven unmixing methods on the Jasper Ridge dataset. (Top to Bottom) Rows are the spectral signatures of tree, water, soil, and road, respectively. Solid lines denote the reference endmembers, and dashed lines denote the estimated endmembers. (a) TV-RSNMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) MNN-BU. (f) UnDIP. (g) SNMF-Net.

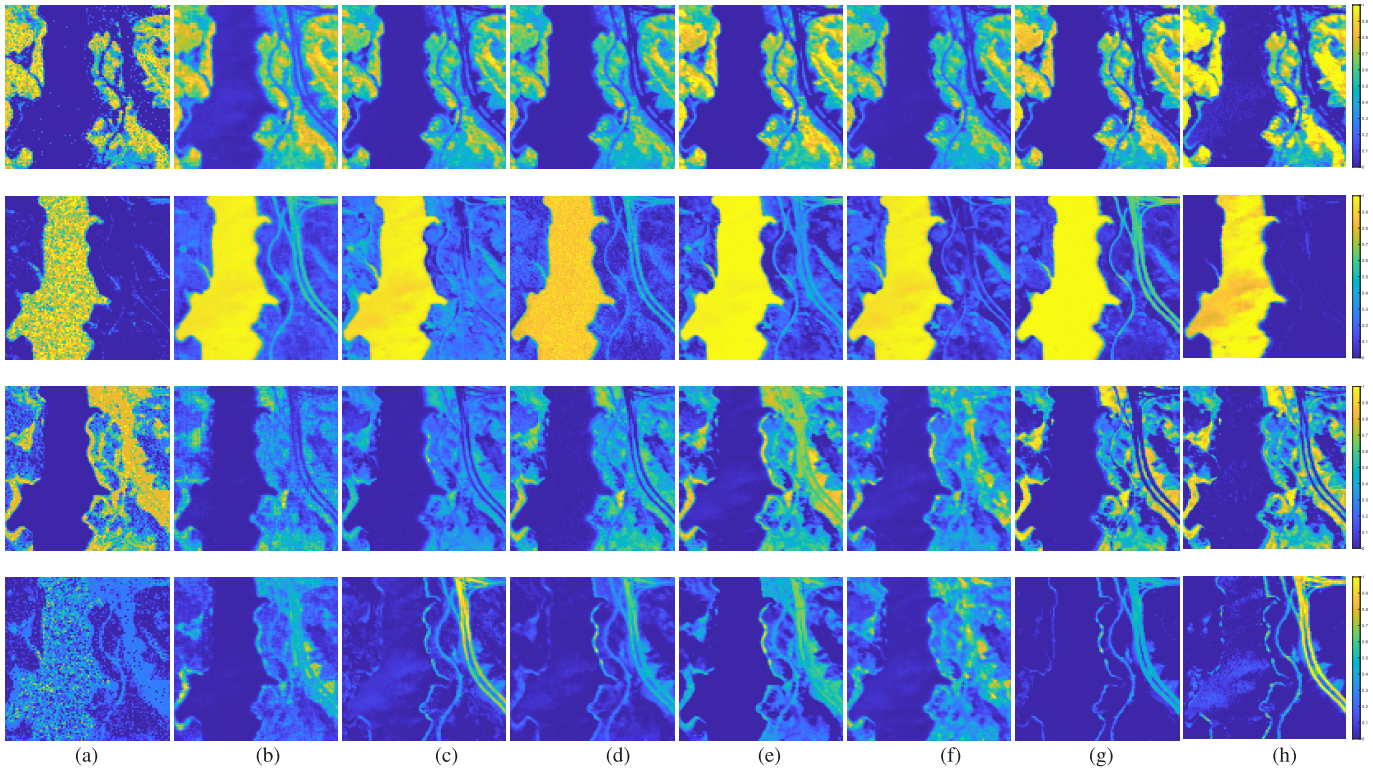


Fig. 11. Estimated abundance maps by seven unmixing methods on the Jasper Ridge dataset. (Top to Bottom) Rows are the abundance maps of tree, water, soil, and road, respectively. (a) TV-RSNMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) MNN-BU. (f) UnDIP. (g) SNMF-Net. (h) Reference.

it can be seen that MV-NTF performs better than TV-RSNMF due to the consideration of the spectral–spatial structure of HSIs powered by tensor algebra. The standard deviations of UnDIP are zero because UnDIP adopts the simplex volume maximization (SiVM) [56] method to extract endmembers, which is insensitive to initializations. Compared with DAEN, learning-based methods with sparsity constraints, i.e., uDAS, MNN-BU, and SNMF-Net, perform better. MNN-BU and SNMF-Net surpass uDAS because they simultaneously take the advantages of the model- and learning-based methods for

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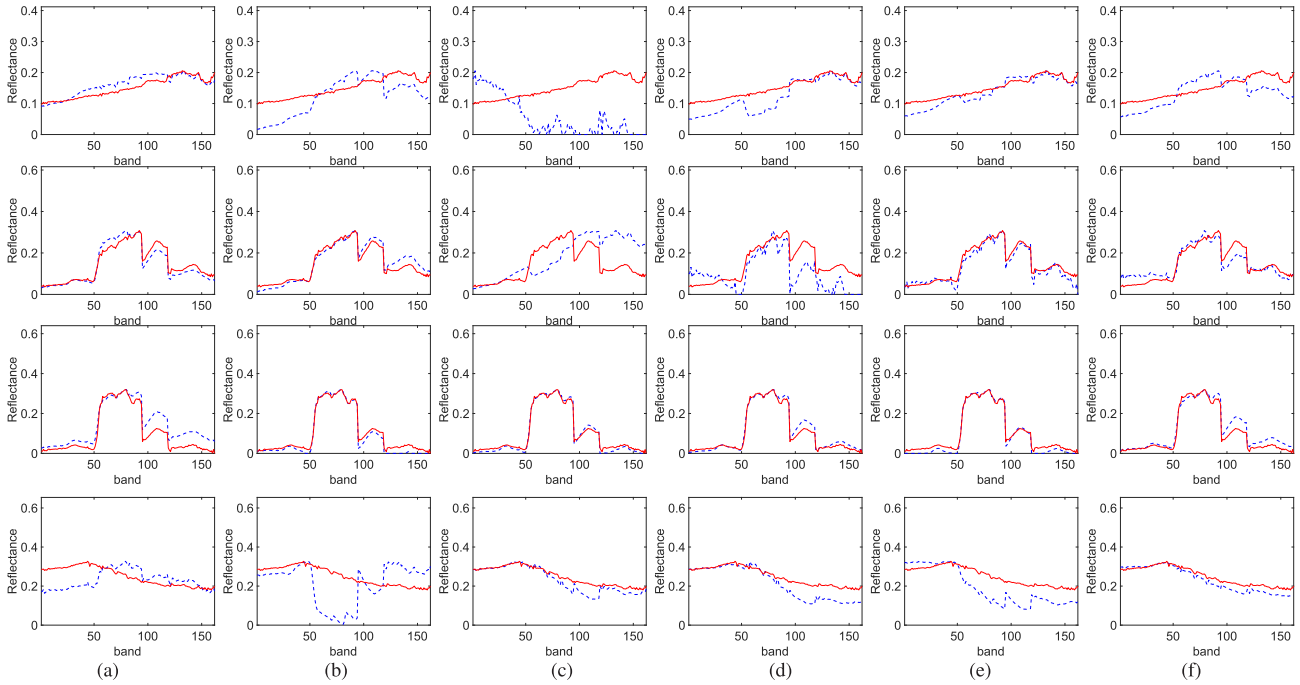


Fig. 12. Estimated endmembers by six unmixing algorithms on the HYDICE Urban dataset. (Top to Bottom) Rows are the spectral signatures of asphalt, grass, tree, and roof, respectively. (a) TV-RSNMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) MNN-BU. (f) SNMF-Net.

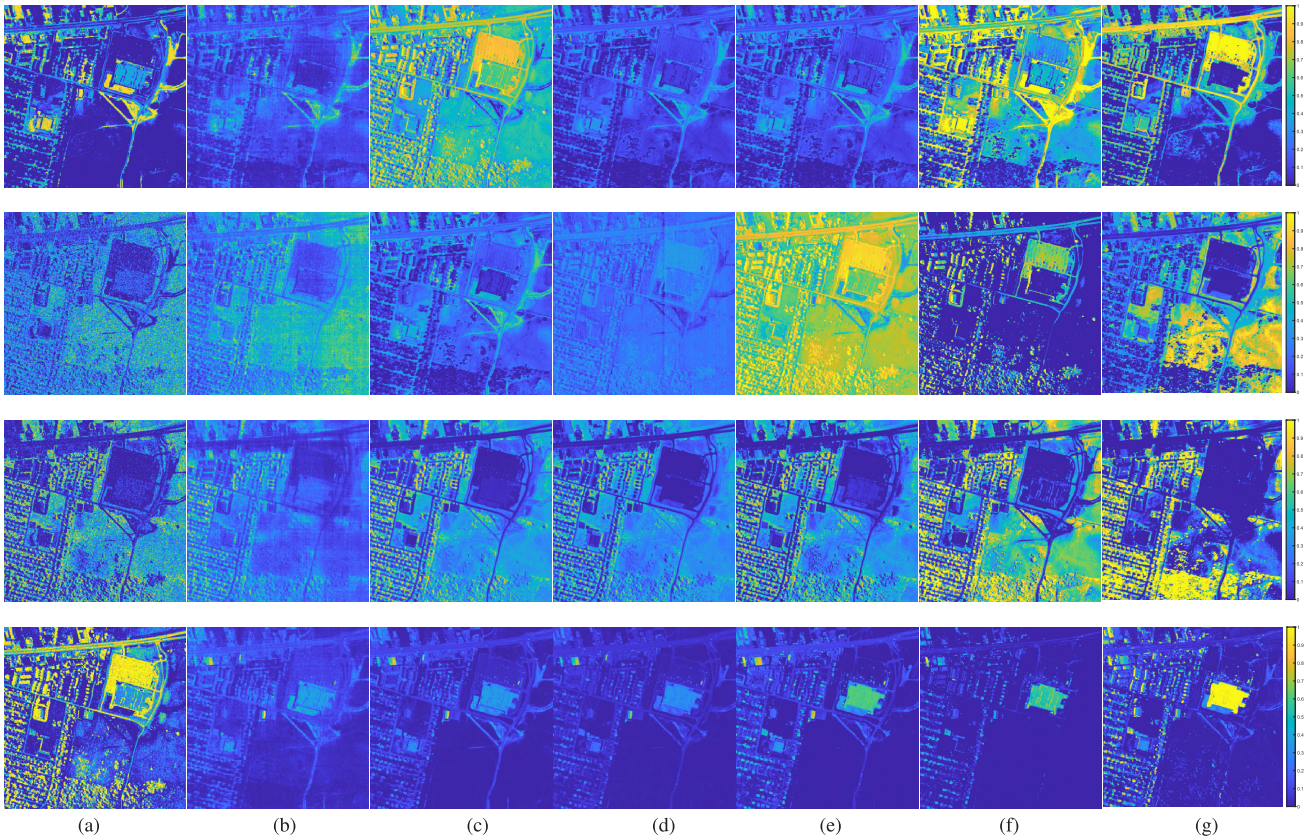


Fig. 13. Estimated abundance maps by six unmixing algorithms on the HYDICE Urban dataset. (Top to Bottom) Rows are the abundance maps of asphalt, grass, tree, and roof, respectively. (a) TV-RSNMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) MNN-BU. (f) SNMF-Net. (g) Reference.

unmixing. Though provided with better performance than two model-based methods, UnDIP provides unsatisfactory SAD than alternative learning-based methods. The main reason is that UnDIP precomputes the endmembers by SiVM and keeps endmembers fixed for abundance estimation. The proposed SNMF-Net stands out among all the methods. On the one

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TABLE II

MEANS AND STANDARD DEVIATIONS OF THE SAD ON URBAN DATA

Algorithm	TV-RSNMF [55]	MV-NTF [18]	DAEN [22]	uDAS [26]	MNN-BU [39]	SNMF-Net
Asphalt	**0.1028±1.17%**	0.1720±0.75%	1.0224±0.27%	0.2174±0.04%	0.1227±1.24%	0.2477±0.93%
Grass	0.1463±1.40%	0.1974±2.97%	0.5122±0.29%	0.6862±33.88%	**0.1234±0.88%**	0.2131±2.81%
Tree	0.2628±1.72%	0.1284±0.32%	**0.0801±0.01%**	0.0955±1.12%	0.1113±0.20%	0.1848±2.05%
Roof	0.3332±8.00%	0.3608±3.66%	0.1069±0.03%	0.2037±2.37%	0.3558±3.80%	**0.0554±2.56%**
Mean	0.2113±1.04%	0.2147±0.96%	0.4304±0.02%	0.3007±8.79%	0.1783±1.22%	**0.1753±0.55%**

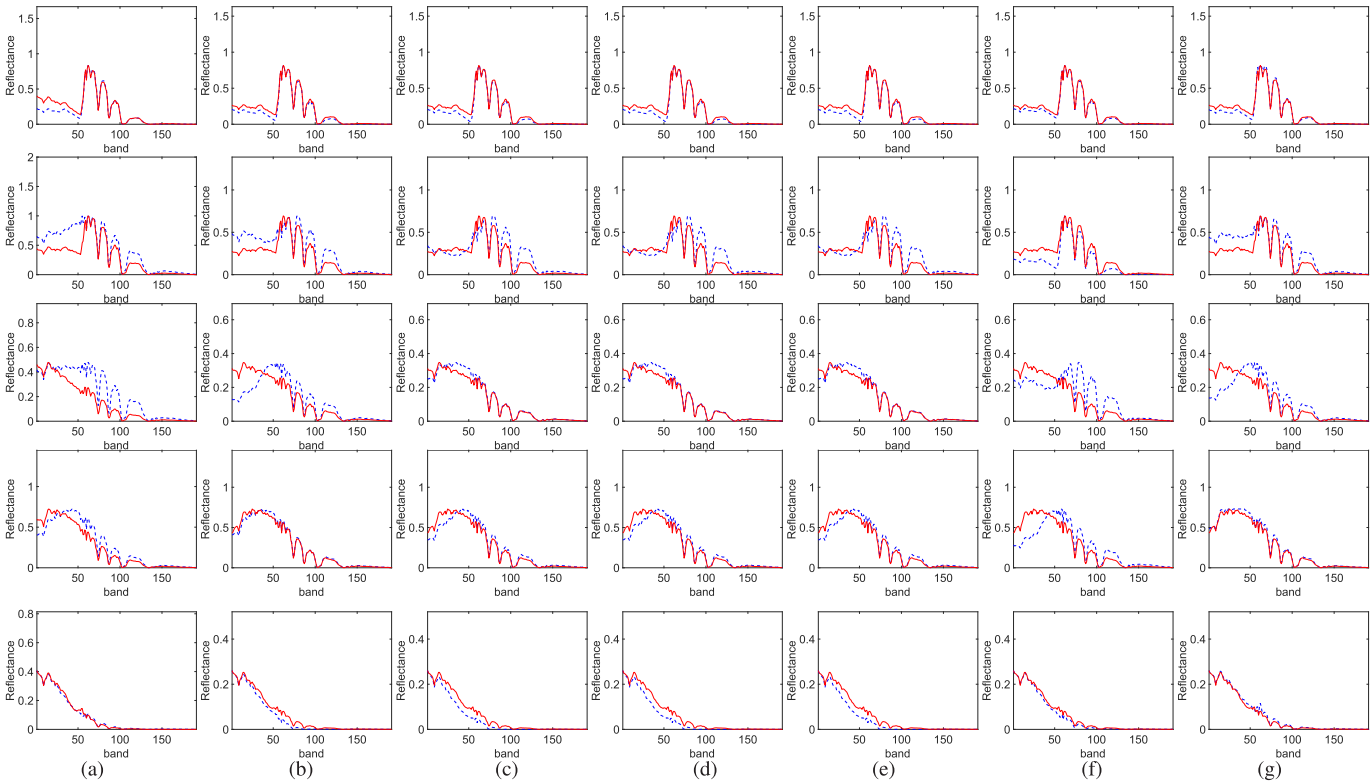


Fig. 14. Estimated endmembers by seven unmixing algorithms on the Washington DC Mall Data. (Top to Bottom) Rows are the spectral signatures of tree, grass, street, roof, and water, respectively. (a) TV-RSNMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) MNN-BU. (f) UnDIP. (g) SNMF-Net.

hand, SNMF-Net adopts alternating network architecture to estimate endmembers and abundances, which fully accords with the alternating optimization of L_p-NMF. On the other hand, SNMF-Net absorbs the spectral mixture model for network constructing and incorporates the prior knowledge on the unmixing problem and sparse representation theory for enhanced learning.

Figs. 10 and 11, respectively, demonstrate the endmembers and abundance map generated by all the methods. The higher brightness (black to white) indicates larger abundance values (0–1). Learning-based approaches produce more attractive abundances because of higher learning capacity. SNMF-Net matches best with reference targets and ranks first among all methods, especially the soil and road target, whereas other methods, such as TV-RSNMF and MV-NTF, have some difficulties in accurate estimation. The experiment further proves the superior unmixing ability of SNMF-Net.

2) HYDICE Urban Dataset: This HSI was acquired by the HYDICE sensor and covers 307 × 307 pixels and 210 bands from 0.4 to 2.5 μm. Urban HSI was corrupted by various noises, which threatens challenges for unmixing. We removed some relative low-SNR bands and water-vapor bands, leaving 162 bands to conduct an experiment whose 80th band is illustrated in Fig. 9(b). We set four material targets, including roof, grass, asphalt road, and tree. The Urban dataset contains outliers and noises, which makes it difficult to extract endmembers accurately. For this reason, UnDIP can only extract one endmember, and we did not include its unmixing result in this experiment.

Table II quantitatively compares the endmembers generated by all the methods in terms of the SAD index. It can be observed that MNN-BU and SNMF-Net provide comparatively better unmixing performance because of the hybrid advantages of learning- and model-based methods. SNMF-Net is slightly superior to MNN-BU using half as many training samples as MNN-BU, showing its better learning ability. Figs. 12 and 13, respectively, give a visual comparison between all the competing methods. In general, all the methods cannot adequately estimate roof and asphalt because of their relatively less number of training samples and the negative effect of outliers and noises. The proposed SNMF-Net dominates other compared methods by sharing more similarities with the reference ones.

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TABLE III
MEANS AND STANDARD DEVIATIONS OF THE SAD ON THE WASHINGTON DC MALL DATA

Algorithm	TV-RSNMF [55]	MV-NTF [18]	DAEN [22]	uDAS [26]	MNN-BU [39]	UnDIP [30]	SNMF-Net
Tree	0.20711±1%	0.1211±0.53%	0.1378±2.33%	0.0939±2.15%	0.1906±0.23%	**0.0795±0%**	0.1309±1.06%
Grass	0.2961±4.34%	0.2357±0.40%	0.3773±17.84%	0.2888±6.03%	0.3513±6.84%	0.2483±0%	**0.2180±2.09%**
Street	0.3338±1.57%	0.4183±0.75%	0.3870±17.49%	0.3270±10.34%	**0.2376±6.46%**	0.4216±0%	0.3836±0.79%
Roof	0.2163±2.84%	0.0818±3.46%	0.1855±10.50%	0.0801±3.06%	0.0915±0.28%	0.3539±0%	**0.0356±0.51%**
Water	**0.0466±0.53%**	0.1317±0.70%	0.1760±5.88%	0.1264±9.28%	0.1544±2.28%	0.0784±0%	0.0602±1.70%
Mean	0.2200±0.51%	0.1977±0.80%	0.2527±3.42%	0.1833±1.99%	0.2051±0.53%	0.2363±0%	**0.1657±0.63%**

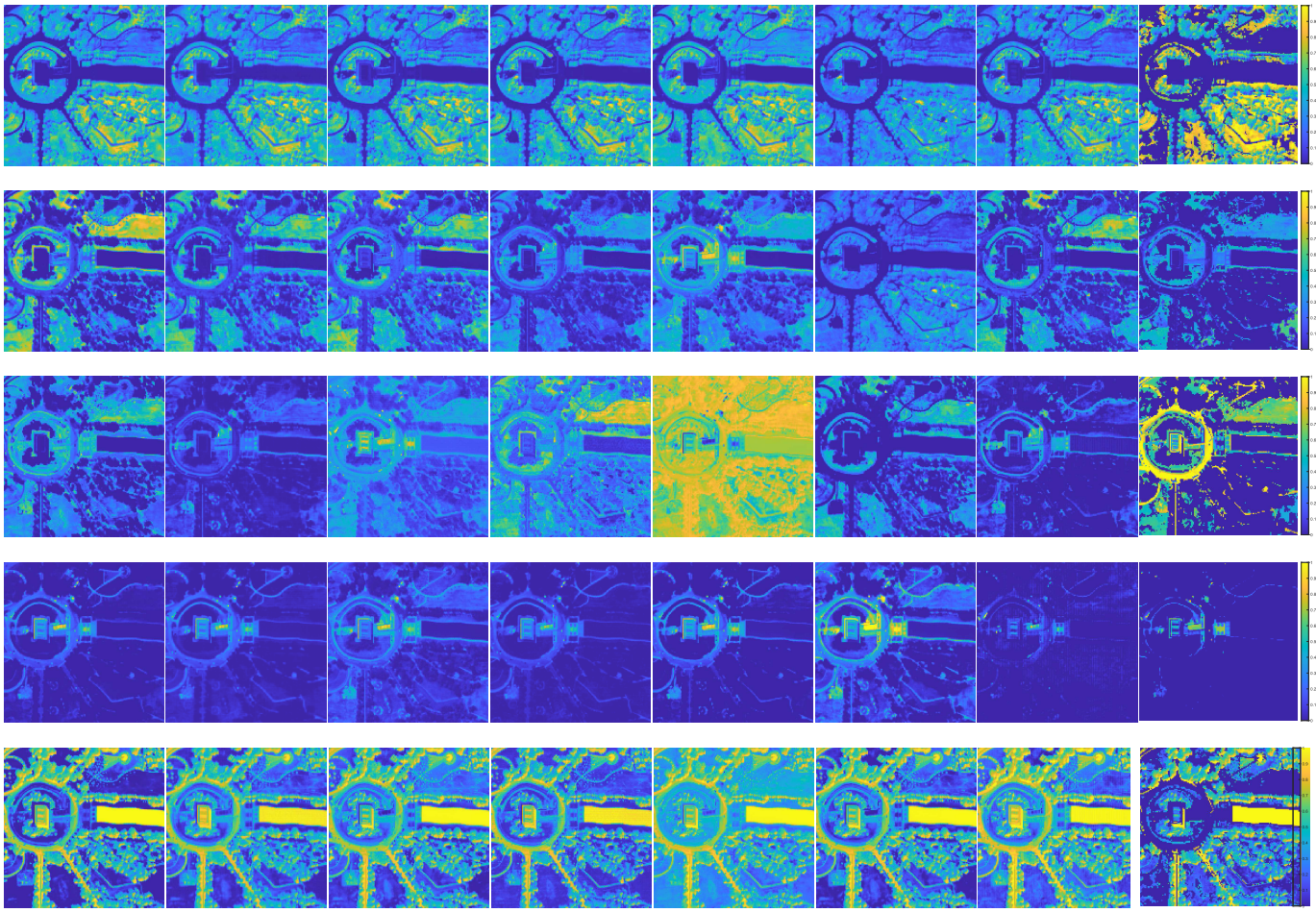


Fig. 15. Estimated abundance maps by seven unmixing algorithms on the HYDICE Urban dataset. (Top to Bottom) Rows are the abundance maps of tree, grass, street, roof, and water, respectively. (a) TV-RSNMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) MNN-BU. (f) UnDIP. (g) SNMF-Net. (h) Reference.

3) Washington DC Mall Dataset: The original Washington DC Mall dataset includes 1208 × 307 pixels and 210 bands of the spectral resolution of 9.52 nm. We used a subimage containing 150 × 150 pixels and 191 bands for experiments by removing the noisy spectral bands (103–106, 138–148, and 207–210). Fig. 9(c) shows the third-band image of the Washington DC Mall dataset. It was assumed that the image provides tree, grass, street, roof, and water materials.

Table III presents a comparable SAD of all the methods. Using only 500 pixels, SNMF-Net outperforms all other competed methods, showing the advantages of absorbing the knowledge of model-based methods into learning-based methods for unmixing. Figs. 14 and 15 describe the endmembers and abundances generated by all the methods. The deep prior learned by the neural networks helps UnDIP achieve more accurate endmembers and abundance maps on the tree and roof target. As the phenomenon revealed in Table III, SNMF-Net performs better in tree, water, and roof target but is inferior to alternative methods in street and grass targets. This may because the spectral signature of the latter two targets are, respectively, very similar to water and tree, making SNMF-Net difficult to distinguish especially using only a subset of all the pixels to train the network parameters.

V. CONCLUSION

In this article, we propose an interpretable network SNMF-Net for unmixing by unrolling the optimization of L_p sparsity constrained NMF unmixing method. SNMF-Net

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equips with the hybrid merits of the model- and learning-based methods. The network architecture of SNMF-Net is consistent with the LMM and the alternating optimization of L_p-NMF. All the parameters are explainable and support end-to-end training. Experimental results on both synthetic and real-world HSI show the advantages of SNMF-Net over many state-of-the-art methods, i.e., higher interpretability, stronger learning ability, less requirement of training samples, and easy integration of domain knowledge. We will incorporate the spatial information into the network for improved unmixing in our further research.

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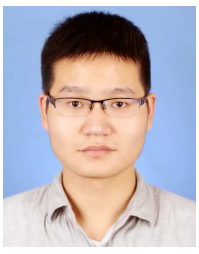
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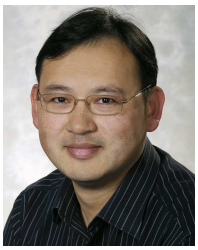
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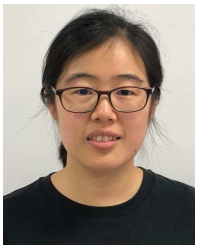
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